**Q1. JEE Main 2026 (21 January Shift 2)**

The first and second ionization constants of $H_2X$ are $2.5 \times 10^{-8}$ and $1.0 \times 10^{-13}$ respectively. The concentration of $X^{2-}$ in $0.1 M H_2X$ solution is ____ $\times 10^{-15}$M. (Nearest Integer)

**Q2. JEE Main 2026 (23 January Shift 1)**

$x$mg of pure HCl was used to make an aqueous solution. 25.0 mL of $0.1 M Ba(OH)_2$ solution is used when the HCl solution was titrated against it. The numerical value of $x$ is ____ $\times 10^{-1}$. (Nearest integer) Given : Molar mass of HCl and $Ba(OH)_2$ are 36.5 and 171.0 g $mol^{-1}$ respectively.

**Q3. JEE Main 2026 (22 January Shift 2)**

Which of the following mixture gives a buffer solution with $pH = 9.25$ ?

Given : $pK_b (NH_4OH) = 4.75$

(1) $0.2 M NH_4OH (0.5 L) + 0.1 M HCl (0.5 L)$

(2) $0.2 M NH_4OH (0.4 L) + 0.1 M HCl (1 L)$

(3) $0.4 M NH_4OH (1 L) + 0.1 M HCl (1 L)$

(4) $0.5 M NH_4OH (0.2 L) + 0.2 M HCl (0.5 L)$

**Q4. JEE Main 2026 (28 January Shift 1)**

Consider a weak base ' B ' of $pK_b = 5.699$. ' $x$ ' mL of 0.02 M HCl and ' y ' mL of 0.02 M weak base ' B ' are mixed to make 100 mL of a buffer of pH 9 at 25°C. The values of ' $x$ ' and ' $y$ ' respectively are: [Given: $\log 2 = 0.3010, \log 3 = 0.4771, \log 5 = 0.699$]

(1)

| $x$ | $y$ |
|---|---|
| 85.7 | 14.3 |

(2)

| $x$ | $y$ |
|---|---|
| 14.3 | 85.7 |

(3)

| $x$ | $y$ |
|---|---|
| 11.1 | 88.9 |

(4)

| $x$ | $y$ |
|---|---|
| 42.7 | 57.3 |

**Q5. JEE Main 2026 (28 January Shift 1)**

Consider the dissociation equilibrium of the following weak acid

$HA \rightleftharpoons H^+(aq) + A^-(aq)$

If the pKa of the acid is 4, then the pH of 10 mMHA solution is ____.(Nearest integer)

[Given: The degree of dissociation can be neglected with respect to unity]

**Q6. JEE Main 2026 (24 January Shift 1)**

Consider two Group IV metal ions $X^{2+}$ and $Y^{2+}$. A solution containing $0.01 M X^{2+}$ and $0.01 M Y^{2+}$ is saturated with $H_2S$. The pH at which the metal sulphide YS will form as a precipitate is ____. (Nearest integer)
(Given: $K_{sp}(XS) = 1 \times 10^{-22}$ at $25°C$, $K_{sp}(YS) = 4 \times 10^{-16}$ at $25°C$, $[H_2S] = 0.1M$ in solution, $K_{a1} \times K_{a2}(H_2S) = 1.0 \times 10^{-21}$, $\log 2 = 0.30$, $\log 3 = 0.48$, $\log 5 = 0.70$)

## ANSWERS AND SOLUTIONS

**1.** 100 **2.** 1825 **3.** (1) **4.** (2) **5.** 3 **6.** 4

**1.** **(100)** For a weak diprotic acid $H_2X$, the ionization steps are:

$H_2X \rightleftharpoons H^+ + HX^-$ with $K_{a1} = 2.5 \times 10^{-8}$

$HX^- \rightleftharpoons H^+ + X^{2-}$ with $K_{a2} = 1.0 \times 10^{-13}$

Since $K_{a1} \gg K_{a2}$, the concentration of $H^+$ is primarily determined by the first ionization.

$[H^+] \approx \sqrt{K_{a1} \cdot C} = \sqrt{2.5 \times 10^{-8} \times 0.1} = \sqrt{2.5 \times 10^{-9}} = 5 \times 10^{-5} M$

Also, from the first ionization, $[HX^-] \approx [H^+] = 5 \times 10^{-5} M$

For the second ionization:

$K_{a2} = \frac{[H^+][X^{2-}]}{[HX^-]}$

Substituting the values: $1.0 \times 10^{-13} = \frac{(5\times10^{-5})[X^{2-}]}{5\times10^{-5}}$

Thus, $[X^{2-}] = K_{a2} = 1.0 \times 10^{-13} M$

To express this in the form $x \times 10^{-15} M$:

$1.0 \times 10^{-13} = 100 \times 10^{-15} M$

The value is 100.

**2.** **(1825)** The reaction between HCl and $Ba(OH)_2$ is:

$2HCl + Ba(OH)_2 \rightarrow BaCl_2 + 2H_2O$

Moles of $Ba(OH)_2$ used: n = 0.1 M × 0.025 L = 0.0025 mol

From stoichiometry: moles of HCl = 2 × 0.0025 = 0.005 mol

Mass of HCl = 0.005 mol × 36.5 g/mol = 0.1825 g = 182.5 mg

Expressing as x = ___ × $10^{-1}$:

182.5 = 1825 × $10^{-1}$

The numerical value is 1825 (or 1.825 × $10^2$ mg when expressed with standard scientific notation)

**3.** **(1)** $pOH = pK_b + \log \frac{Salt}{Base}$

$4.75 = 4.75 + \log \frac{Salt}{Base}$

Milimoles of [Salt] = milimoles of [Base]

$NH_4OH + HCl \rightarrow NH_4Cl + H_2O$

$0.2M, 0.5L \quad 0.1M, 0.5L$

100 mmole 50 mmole

50 mmole – 50 mmole

Milimoles of $NH_4OH$ = milimoles of $NH_4Cl$

**4.**

| | HCl | + B | → | $BH^+ + Cl^-$ |
|---|---|---|---|---|
| | 0.02M x ml | 0.02M y ml | | |
| $t_f$ | 0 | 0.02y–0.02x | | 0.02x |

**(2)**

$$pOH = pK_b + \log\left[\frac{Salt}{Base}\right]$$

$$5 = 5.699 + \log\left[\frac{Salt}{Base}\right]$$

$$\frac{x}{y-x} = \frac{1}{5}$$

$$6x = y$$

$$7x = 100$$

$$x = \frac{100}{7}\text{ ml}$$

$$y = \frac{600}{7}\text{ ml}$$

**5.** **(3)** For the weak acid dissociation $HA \rightleftharpoons H^+ + A^-$ with pKa = 4

We have Ka = $10^{-4}$.

For a 10 mM (0.01 M) solution where the degree of dissociation is negligible with respect to unity, the concentration of HA remains approximately 0.01 M.

The equilibrium expression gives

$$Ka = \frac{[H^+]^2}{[HA]}$$

Substituting: $10^{-4} = \frac{[H^+]^2}{0.01}$, which yields $[H^+]^2 = 10^{-6}$, so $[H^+] = 10^{-3}$ M.

Therefore, pH = $-\log(10^{-3}) = 3$.

**6.** **(4)** $XS(s) \rightleftharpoons X^{+2}(aq.) + S^{2-}(aq.)$

For precipitation of $XS(s)$

$$[X^{+2}][S^{2-}] \geq K_{sp}(XS)$$

$$[S^{2-}] \geq \frac{1 \times 10^{-22}}{0.01} = 10^{-20}$$

$$YS(s) \rightleftharpoons Y^{+2}(aq) + S^{2-}(aq)$$

For precipitation of $YS(s)$

$$[Y^{+2}][S^{2-}] \geq K_{sp}(YS)$$

$$[S^{2-}] \geq \frac{4 \times 10^{-16}}{10^{-2}} = 4 \times 10^{-14}$$

Now, $H_2S(aq) \rightleftharpoons 2H^+(aq) + S^{2-}(aq)$

$$\frac{[S^{2-}][H^+]^2}{H_2S} = K_{a_1} \times K_{a_2} = 1 \times 10^{-21}$$

$$[S^{2-}] = \frac{1 \times 10^{-21} \times [H_2S]}{[H^+]^2} \geq 4 \times 10^{-14}$$

$$[H^+]^2 \leq \frac{1}{4} \times 10^{-7} \times 10^{-1}$$

$[H^+] \leq \frac{1}{2} \times 10^{-4} \Rightarrow pH \geq 4.3$

Nearest integer pH = 4.